



Regression & Time Series

- Regression
- Forecasting

Regression



We speak of **regression** when the target variable to be predicted/forecast is a **real-valued variable**

Examples:

- Predict salary as a function of employee features
- Forecast stock price given historical data

Regression



Classic approaches

- **KNN-regression**
- **Least-squares linear regression**
- Neural networks
- Regression trees
- ...

Non-Parametric Regression



Parametric Regression

- Choosing **in advance** the **shape** of the function
- E.g., Linear regression

Non-Parametric Regression

- **No prior choice** on the **shape** of the function
- E.g., KNN-regression

Regression



KNN-regression

- An extension of KNN algorithm to **regression** case
- Simply compute target value as (weighted) **combination** of those of the K nearest neighbors

$$\hat{y}_j = \frac{\sum_{k \in \mathcal{N}_j} w_k y_k}{\sum_{k \in \mathcal{N}_j} w_k}$$

$$w_k = \frac{1}{d(x_j, x_k)}$$

Linear Regression



Linear regression with **least-squares** approach

Minimize the **Residual Sum of Squares** (RSS)

$$RSS = \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

where $\hat{y}_i$ is computed as

$$\hat{y}_i = \beta^T x = \sum_{i=1}^K \beta_i x_i$$

Linear Regression



In matrix form we are given N examples with K features: $Y \in \mathbb{R}^N, \beta \in \mathbb{R}^K, X \in \mathbb{R}^{N \times K}$

$$\min_{\beta} \|Y - \beta^T X\|^2$$

Linear Regression



The problem is **over-dimensional**: we have more equations than variables ($N \gg K$)...

There exist a closed-form solution

$$N \gg K$$

$$\beta^* = \underbrace{(X^T X)^{-1} X^T}_{\substack{\text{MOORE-PENROSE} \\ \text{PSEUDO-INVERSE}}} Y$$

Linear vs. Non-Linear

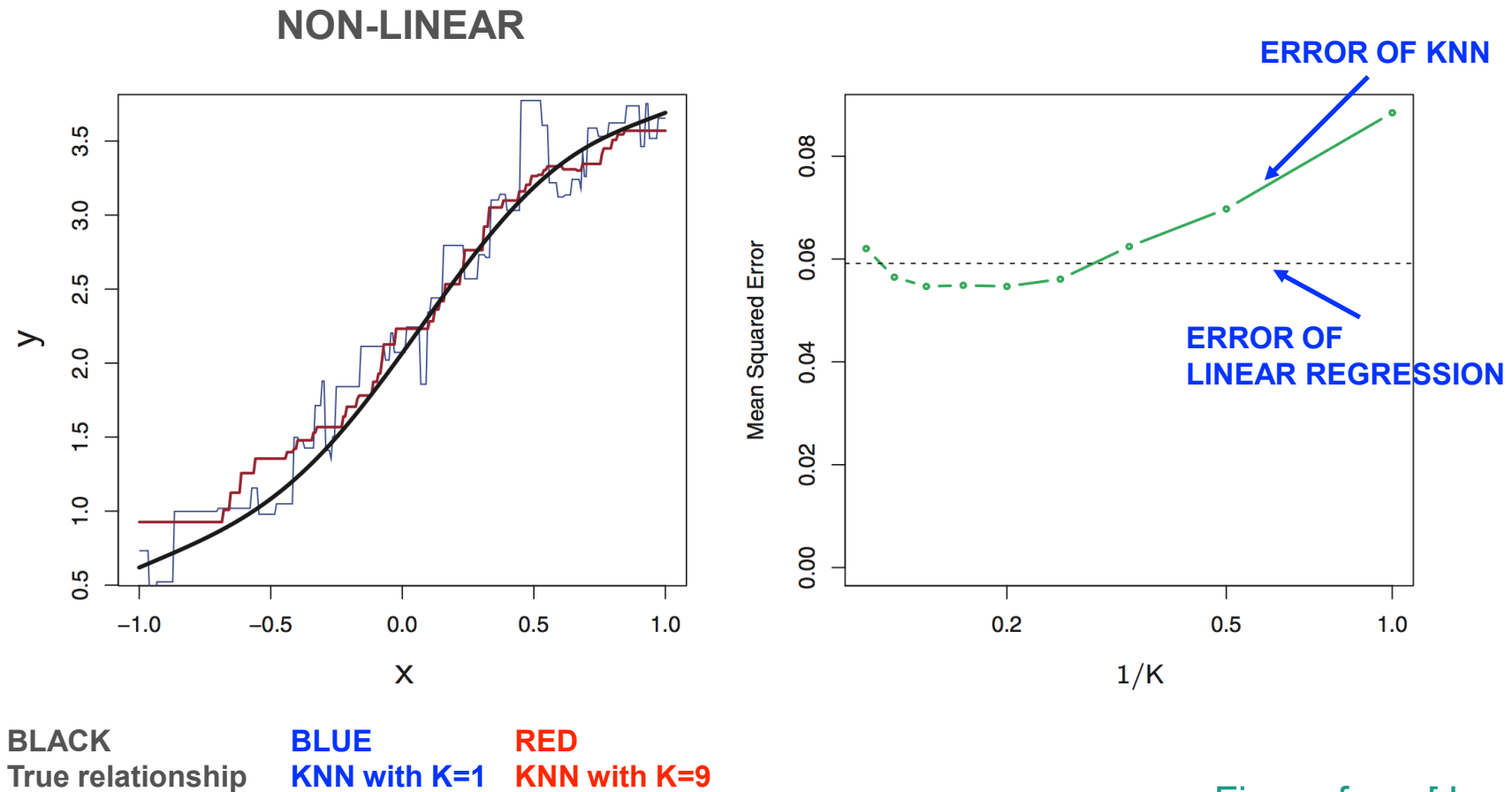
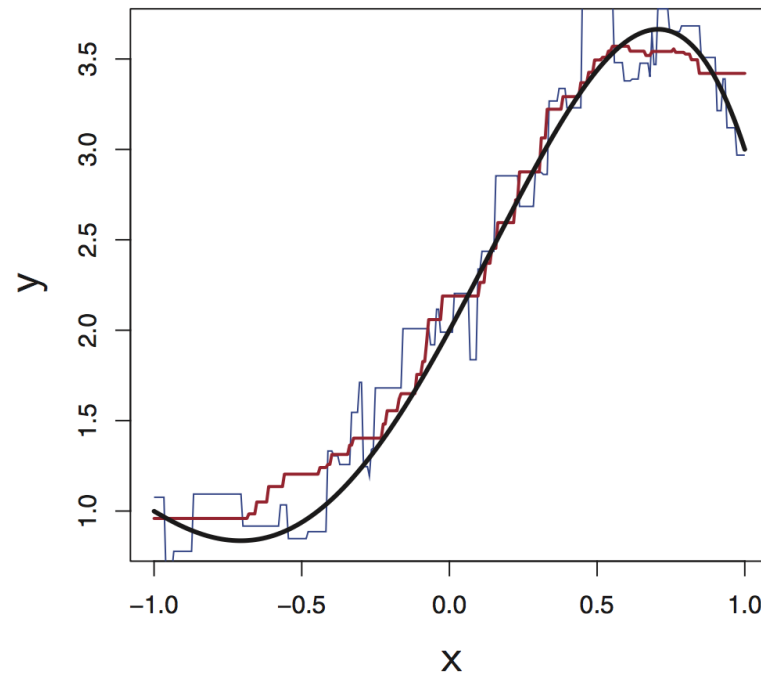


Figure from [James et al.]

Linear vs. Non-Linear

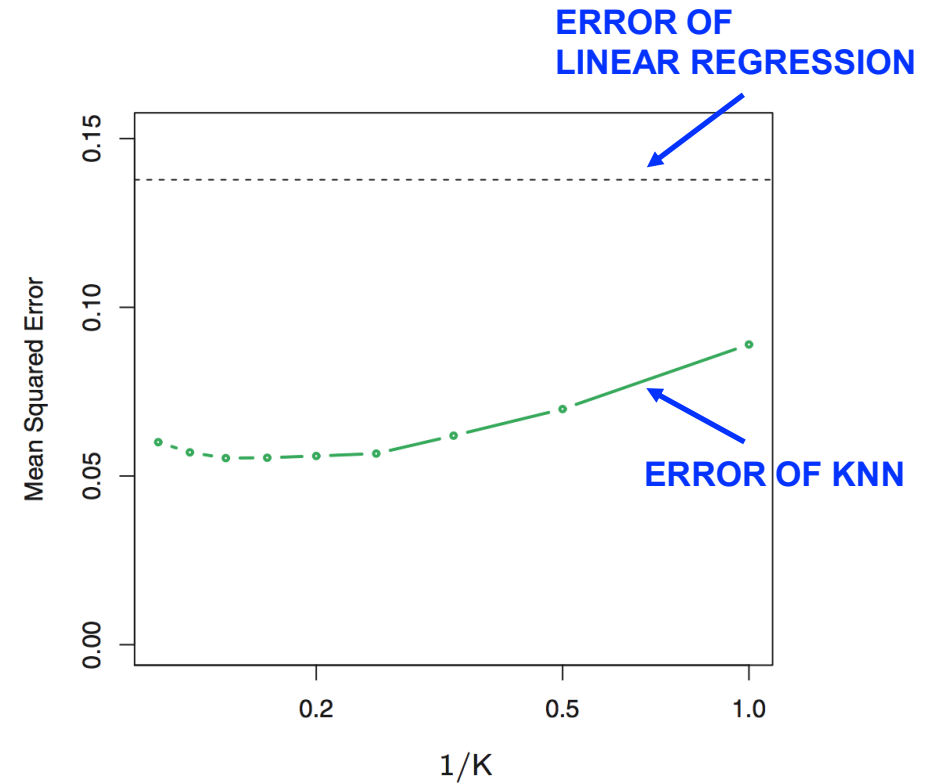
STRONGLY NON-LINEAR



BLACK
True relationship

BLUE
KNN with $K=1$

RED
KNN with $K=9$



ERROR OF
LINEAR REGRESSION

ERROR OF KNN

Figure from [James et al.]

More on Regression



Other regression models:

- Regression trees
- Neural networks
- ...

Regression Trees



An **extension** of decision trees to regression

- Use the **improvement** in Mean Squared Error to evaluate and select attributes during training
- In a leaf node, compute the **average** of the target variable across the training examples in the leaf
- Use **ensemble** techniques (Random Forests and Gradient Boosting) as for classification

Neural networks



Any neural architecture can be used for regression

- For Multi-Layer Perceptron, just use **one neuron** in the output layer, with linear activation function
- Use a **loss function** that evaluates regression tasks, such as Mean Squared (or Absolute) Error
- Specific architectures are dedicated to time-series analysis: i.e., **recurrent neural networks** such as Long Short-Term Memory (LSTM) networks

Example on Time-Series

Parameters to control when forecasting time-series?



DATA AGGREGATION?

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- <https://medium.com/data-science/how-not-to-use-machine-learning-for-time-series-forecasting-avoiding-the-pitfalls-19f9d7adf424>